A matrix representation of the imaginary unit

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1 Introduction

I think that, on the surface at least, it is fair to bring into question the existence of a matrix representation of the imaginary unit, i. But logically it does follow. We're quite used to using matrices to represent rotations and reflections. We're also used to i representing a $90°$ rotation. But, in this little document, I want to draw up a more interesting comparison between these two objects, particularly through a group theoretic perspective.

2 Our two groups

Let us first consider two prototypical examples of Lie groups. The first of which is $SO(2)$, the group of 2×2 orthogonal matrices with a unit determinant, often written as:

$$\begin{pmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{pmatrix} \in SO(2) \tag{1}$$

You may recognise that this looks a lot like a rotation matrix, and that is because it is! $SO(2)$ is typically viewed as the group of 2d rotations.

Our second group is $U(1)$, the group of all unitary 1×1 matrices, often written as:

$$e^{i\theta} \in U(1) \tag{2}$$

The astute amongst you may have realised that these two objects are in fact the same. They can both be used to represent a 2d rotation. Because of this fact, we can identify an isomorphism (iso meaning "equal" or "same" and morphism meaning "structure preserving map") between them:

$$SO(2) \cong U(1) \tag{3}$$

3 The Lie algebra

One important part of the study and subsequent usage of Lie groups is the fact that they double as manifolds. One way to then define the Lie algebra of a Lie group (often denoted as the group's name written in Fraktur) is as the tangent space to the group evaluated at the identity element. That is to say:

$$\left.\frac{\partial G}{\partial \theta_i}\right|_{\mathbb{I}} \in \mathfrak{g} \tag{4}$$

Let us first compute an element of the Lie algebra of $U(1)$, $\mathfrak{u}(1)$:

$$\frac{\partial}{\partial \theta}\left[e^{i\theta}\right]_{\theta=0} = i \tag{5}$$

Doing the same for $SO(2)$ yields:

$$J = \frac{\partial}{\partial \theta}\begin{pmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{pmatrix}_{\theta=0} = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix} \tag{6}$$

So here, we are presented with a matrix representation of the imaginary unit, i. We can see that this matrix does literally represent an anti-clockwise $90°$ rotation. Let us also see if it obeys $i^2 = -1$:

$$\begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}\begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix} = \begin{pmatrix} -1 & 0 \\ 0 & -1 \end{pmatrix} = -\mathbb{I}_2 \tag{7}$$

Indeed, the matrix squares to the negative 2×2 identity matrix, just as i squares to the negative identity element of the field of complex numbers. Let us also see if Euler's identity:

$$e^{i\theta} = \cos\theta + i\sin\theta \tag{8}$$

Is satisfied. You may not be used to exponentiating matrices, but it is quite easy if we work in the Taylor series:

$$\exp\begin{pmatrix} 0 & -\theta \\ \theta & 0 \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} + \theta\begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix} + \frac{\theta^2}{2}\begin{pmatrix} -1 & 0 \\ 0 & -1 \end{pmatrix} + \frac{\theta^3}{3!}\begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix} + ... \tag{9}$$

We begin to see a pattern here, the Taylor series for cosine is appearing in the main diagonal, whilst antisymmetric sines are appearing off diagonal:

$$\exp\begin{pmatrix} 0 & -\theta \\ \theta & 0 \end{pmatrix} = \begin{pmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{pmatrix} \tag{10}$$

You may have now noticed that this is again the prototypical element of $SO(2)$, we have now discovered another definition of the Lie algebra:

$$\mathfrak{g} = \{g : e^{\theta g} \in G \ \forall \theta \in \mathbb{R}\} \tag{11}$$

That is, we can map back to the associated Lie group by exponentiating the Lie algebra element.

With that being said, we still have not found whether Euler's identity is fully satisfied. We can expand (9) in another way, we see that the cosines lie on the diagonal and the sines exist on a term that is exactly $\sin\theta$ lots of J, therefore, we can write:

$$e^{J\theta} = \cos(\theta)\mathbb{I}_2 + \sin(\theta)J \tag{12}$$

Which is then exactly Euler's identity in matrix form.